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Of Mice and Gamers

Gaming mice have started advertising insane specifications but do you really need that much?

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“I simply cannot tolerate my mouse going below 3,600 DPI” – Random Guy on the Train.

My head immediately turned to see who had uttered that phrase, a phrase that just reeked of ignorance. I’m pretty sure that Douglas Engelbart probably twitched in his grave each time something akin was said. I had to get down at the next stop so spinning up a conversation about mice with a complete stranger was out of the question. So here we are, about to dispel a few misconceptions about mice.

DPI or CPI or PPI? There have been numerous phrases to describe sensor sensitivity through the ages. The correct term is CPI (Counts Per Inch) and it refers to the number of steps a sensor will report when it moves one inch. Dots Per Inch, pretty much refers to the same thing but isn’t technically correct. The same can be said of Pulses Per Inch (PPI) when optical sensors were first implemented. Jargon aside, this metric simply tells you how sensitive the mouse sensor happens to be. So a 12,000 DPI mouse simply means that the sensor can generate a lot more data – unwanted data – that can be used to produce motion. This raw data is then processed to produce whatever DPI that the software has set. Here’s what the operating system uses to move the mouse cursor around the screen:

Movement(vertical/horizontal) = mouse.sample(dt, dpi) x sensitivity

Essentially what most people do, is that they ramp up DPI and drop sensitivity to get accurate tracking. So what remains to be seen is the sweet spot that is favourable for the player. Most professional players hardly ever go above 800 DPI. So those ultra-super-premium-holy-lord-of-gaming mice with 12,000 DPI are not favoured by professional players. In fact, most people prefer reducing the sensitivity to such an extent that they like moving their mice across a greater surface. So while they may navigate the usual OS window at 3,600 DPI with the mouse hardly ever moving an inch from a spot, they’ll be all over the mouse mat while gaming.

Another idiotic thing is mouse acceleration. Basically, the faster you flick the mouse the faster your cursor gets. So if you were to flick the mouse an inch to the right very slowly, you’ll trace an inch, but if you flick faster, your cursor will probably move 2-3 inches. Most professional players turn off mouse acceleration.

And then there’s angle smoothing or just plain smoothing. The way a mouse tracks, it compares pictures of the surface to figure out which direction the mouse moved. Now, depending on what kind of algorithm comes into play, there will be certain moments when the algorithm gets confused and tracks in the wrong direction. This manifests as a minor blip, let’s say you were tracking your mouse horizontally in CS:GO and you notice that your cursor moved up a little, by about a pixel or two. This is a blip or a ripple. Angle smoothing, disregards these blips. So you might be wondering, isn’t that a good thing? Well, if you are using Photoshop to trace something, it is actually beneficial. But in game, let’s say you are sniping your enemy and you want to move your cursor down by just one pixel to hit him square in the head. Angle smoothing will consider that one pixel to be an anomaly and will not move your cursor vertically. The sheer frustration that ensues can hardly be described.

Lastly, there is the much debated issue of laser vs optical sensors. Firstly, the two sensors are both optical CMOS sensors if we are to be technically correct. It’s that they are illuminated differently, LED illuminated mice don’t pick up much detail but laser goes all the way and gives you much higher resolution. So if the surface is smooth, laser works much better, but the moment the surface gets soft, the laser picks up a lot more of the undulations. This is unwanted information which when processed would result in a lot of blips and ripples that the sensor then skips over, resulting in “acceleration”. Acceleration is the incorrect term, hence the quotes, this is actually an error with image resolution. This is why LED illuminated mice track much better than laser illuminated mice. Also, the error in optical mice is less than one per cent.